Handout of development of the blood vessels and fetal circulation

Block CVS

2nd year

Prepared by

Prof. Dr. Faten Youssif Mahmoud
Faculty of Medicine –Assiut University

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Development of blood vessels

Aortic arch derivatives

Pharyngeal arches and their aortic arches appear in a craniocaudal sequence. Aortic arches arise from the aortic sac and terminate in right and left dorsal aortae. Caudal to the pharyngeal arches the dorsal aortae fuse to form single descending aorta. When the branchial arches disappear, aortic arches disappear or leave certain arteries.

Fate of aortic arches

Aortic sac -Proximal part of aortic arch.

Dorsal aortae

-Left dorsal aorta -Distal part of aortic arch.

-Right dorsal aorta -Distal part of right subclavian artery.

First aortic arches -Disappear leaving maxillary artery.

Second aortic arches -Disappear leaving stapedial and hyoid arteries.

Third aortic arches -Form carotid arteries.

Fourth aortic arches

-Left 4th aortic arch -Form middle part of aortic arch.

-Right 4th aortic arch -Form proximal part of right subclavian artery.

Fifth aortic arches (disappear)

Sixth aortic arches

-Left 6th aortic arch -Gives left pulmonary artery and ductus arteriosus.

-Right 6th aortic arch -Gives right pulmonary artery.

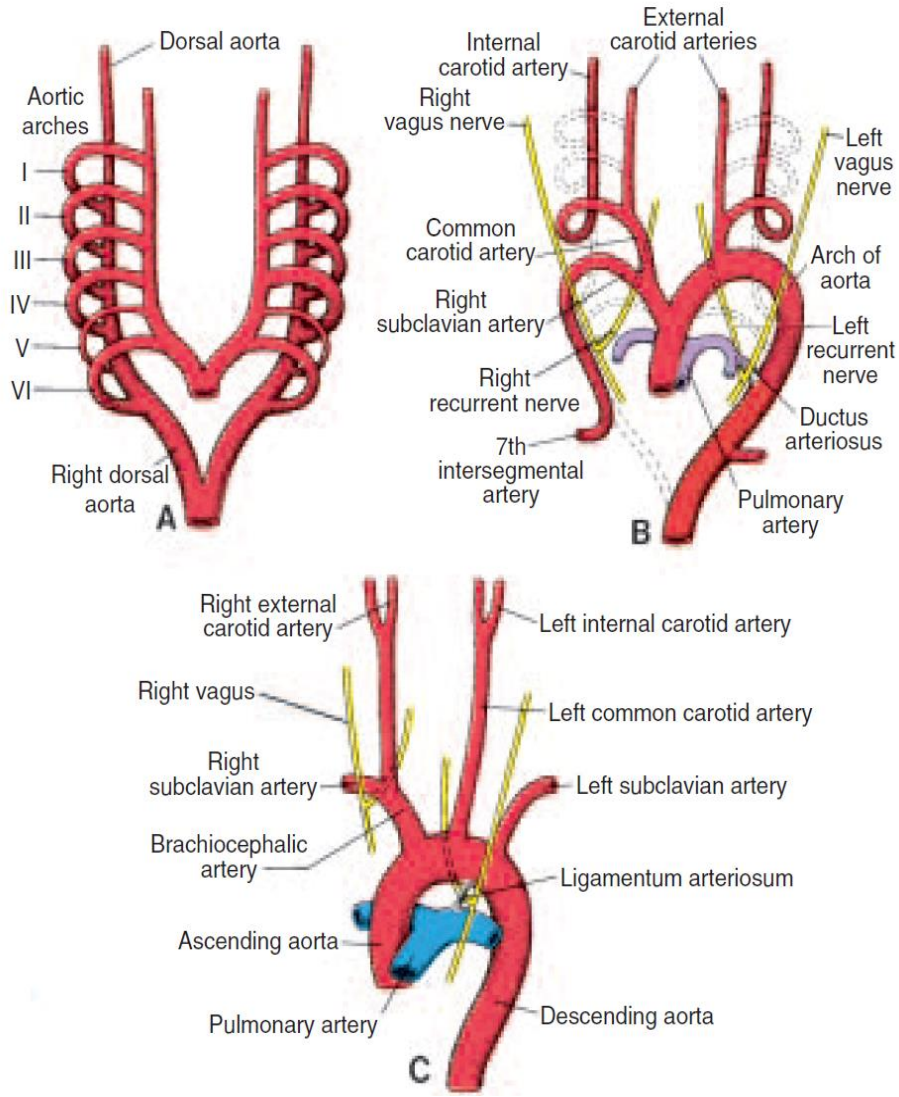


Figure 11.35 A. Aortic arches and dorsal aortae before transformation into the definitive vascular pattern. **B.** Aortic arches and dorsal aortae after the transformation. *Broken lines*, obliterated components. Note the patent ductus arteriosus and position of the seventh intersegmental artery on the left. **C.** The great arteries in the adult. Compare the distance between the place of origin of the left common carotid artery and the left subclavian in **B** and **C.** After disappearance of the distal part of the sixth aortic arch (the fifth arches never form completely), the right recurrent laryngeal nerve hooks around the right subclavian artery. On the left the nerve remains in place and hooks around the ligamentum arteriosum.

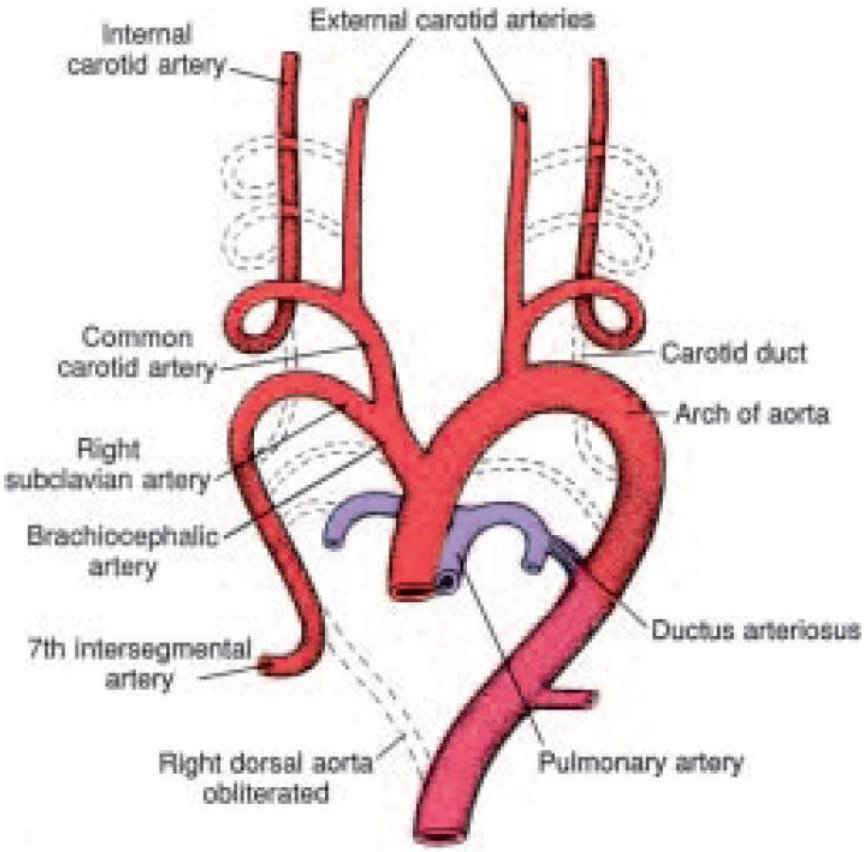


Figure 11.36 Changes from the original aortic arch system.

Branches of dorsal aortae and their fate

Dorsal aorta develops *ventral, lateral and posterolateral branches.*

1-Ventral branches of dorsal aorta

a)Vitelline arteries are paired arteries fuse and form single arteries in dorsal mesentery of the gut**:**

i-Celiac trunk to the foregut.

ii-Superior mesenteric artery to the midgut.

iii-Inferior mesenteric artery to the hindgut.

b)Umbilical arteries, the proximal part gives superior vesical and internal iliac arteries, the distal part obliterates to form medial umbilical ligament.

2-Lateral splanchnic arteries supply intermediate mesoderm. They give renal, suprarenal, phrenic and gonadal arteries.

3-Posterolateral branches are somatic branches. They are ***30 pairs of dorsal inter-segmental* arteries** that pass in between and carry blood to somites and their derivatives. Most of original connections of intersegmental arteries to dorsal aorta disappear:

-In the neck they join to form the **vertebral, ascending cervical, deep cervical, superior intercostals, and subclavian arteries**.

-In the thorax they persist as **posterior intercostals arteries.**

-In the abdomen they remain as **lumbar arteries**. The 5th pair of lumbar inter-segmental arteries forms the common iliac arteries.

-In the pelvis they form the **lateral sacral arteries.**

The caudal end of the dorsal aortae becomes the **median sacral artery**

Anomalies of aortic arches

1-Coarctation (constriction) of the aorta

Coarctation is narrowing of the aorta. It is an abnormality of the aortic media followed by intimal proliferation. It is due to genetic or environmental factors.

Symptoms and signs of coarctation of the aorta

Patients suffer from hypertension in upper body and hypotension in lower body. Legs and feet are pale and cold with weak pulse. There is headache and signs of congestive heart failure. Costal notches are prominent in the ribs as appearing in X-ray. Diagnosis is made by radiography, ultrasonography, CAT scan and MRI.

Types:

i-Preductal (infant type): the constriction is proximal to DA. DA usually remains open. Blood flows through DA to descending aorta. Closure of DA at birth results in rapid deterioration of the infant. Prostaglandin infusion is given in an attempt to reopen DA.

ii-Postductal (adult type): is more common type. DA is usually closed. The constriction is distal to DA. This permits development of collateral circulation between intercostal arteries, internal thoracic arteries and descending thoracic aorta to assist passage of blood into the lower parts of the body.

2-Double aortic arch

Mechanism is failure of distal part of right dorsal aorta to disappear.

3-Right aortic arch

When the entire right dorsal aorta persists and the distal left dorsal aorta involutes.

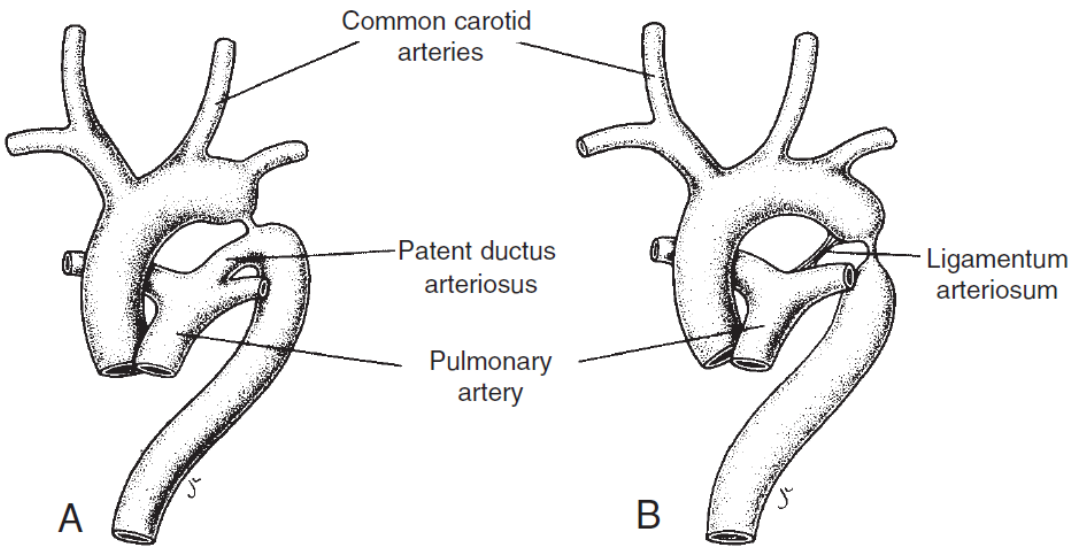


Figure 11.37 Coarctation of the aorta. **A.** Preductal type. **B.** Postductal type. The caudal part of the body is supplied by large hypertrophied intercostal and internal thoracic arteries.

Development of veins

Fate of vitelline veins

Vitelline veins share in formation of:

a)Liver sinusoids. **b)Hepatic veins.**

c)Portal vein. **d)Inferior vena cava.**

Fate of umbilical veins

As the liver develops, umbilical veins lose connection with the heart and share in forming **liver sinusoids.** Right umbilical vein disappears**.**

Fate of cardinal veins

1-The ***anterior cardinal veins*** connect by anastomosis which shunts blood from the left to the right. This shunt becomes the **left brachiocephalic vein.** The cranial portions of the anterior cardinal veins form the **internal jugular veins.** The **Superior vena cava** forms from the right anterior cardinal vein and the right common cardinal vein.

2-The ***posterior cardinal veins*** disappear.

Other cardinal veins form part of IVC.

Development of superior vena cava (SVC)

SVC develops from the right anterior cardinal vein and the right common cardinal vein.

Development of inferior vena cava (IVC)

-Hepatic segment from cranial part of right vitelline vein.

-Renal segment from the right subcardinal vein.

-Sacrocardinal segment from the right sacrocardinal vein.

Anomalies of venae cavae

1-Double superior vena cava.

2-Left superior vena cava.

3-Double inferior vena cava.

4-Absence of hepatic segment of IVC.

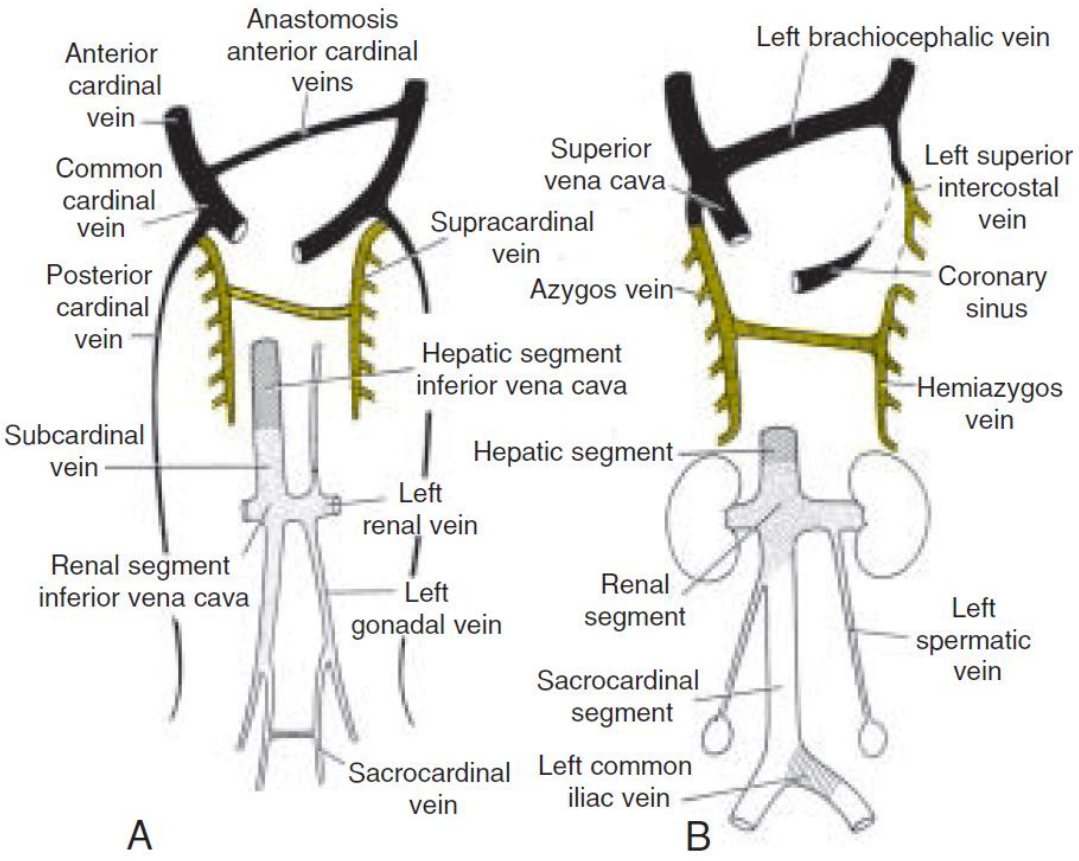


Figure 11.44 Development of the inferior vena cava, azygos vein, and superior vena cava. **A.** Seventh week. The anastomosis lies between the subcardinals, supracardinals, sacrocardinals, and anterior cardinals. **B.** The venous system at birth showing the three components of the inferior vena cava.

Fetal circulation

The blood is circulated into the fetus through a special pathways.

Arterial blood circulation:

The oxygenated blood is conveyed from the placenta **by the umbilical vein.**

This vein enters the abdomen through the umbilical ring and ascends upwards

Along the midline then turn posteriorly along the margin of the faciform ligament for reaching the liver, there it opens into the left branch of the portal vein. This is surrounded by a controllable sphincter. The left branch of the portal vein also communicates with the inferior vena cava(I.V.C.) within the liver by a communicating duct, termed **the ductus venosus.**

Through the I.V.C. to the right and left atrium

The blood passes into the right atrium,. The inferior vena cava opens by a valve which direct the oxygenated blood towards the interatria septum. As the septum is perforated by the foramen oval, thus the oxygenated blood is directly transported under a high pressure into the left atrium.

From the left atrium, the oxygenated blood flows to the left ventricle where a considerable amount of this blood is pumped into the ascending and arch of aorta to supply the brain (through its carotid branches) and the upper limbs (through its subclavian branches) while a small amount of this oxygenated blood travels through the descending aorta to supply the lower limbs.

Venous blood circulation

Since all venous pressure are relatively low, thus the non- oxygenated blood is returned back from the lower limbs and lower part f the body trough the I.V. C.to enter the right atrium but does not pass into the foramen oval. Inststead, it passes into the right ventricle. Similarly, the venous blood is returned from the upper part of the body by the superior vena cava (S.V.C.) to the right atrium and directly to the right ventricle .

On accumulation of the venous blood into the right ventricle, it is pumped into the pulmonary trunk which bifuricates into right and left pulmonary arteries (one to each lung) but because the 2 lungs are still not inflated by air, they have a positive pressure

, **thus the venous blood will be faced by a resistance that shunted it into a communicating duct(ductus arteriosus) between the left pulmonary artery and the aortic arch.** The venous blood continue into the descending aorta which bifurcates into

Two umbilical arteries that running towards the placenta in order to purify this venous blood.

The above explanation indicate that:

1- The inferior vena cava carries both arterial and venous blood, however they are not mixed with each other. This is explained by an opinion that during the venous return inti=o the inferior vena cava, a sphincter will close the opening of the umbilical vein preventing its arterial blood coming from the placenta to be mixed with the venous blood coming from the lower part of the body. Once the flow of the venous stream is completed, the sphincter relaxes with rushing of the accumulated blood into the inferior vena cava (under an increased pressure that built up while the sphincter was closed) and directly reaches to the left atrium

through the foramen oval. Therefore, the oxgen content of the blood in the I.V.C. vary considerably from moment to moment.

2- The aorta carries both arterial and venous blood but they are mixed with each other particularly in the descending aorta.

3- By the time of birth, an immediate post- natal changes will be manifested :

a- The first breathing of the newly born forces the air into the lungs and hence a negative intra-thoracic pressure which causes rushing of the venous blood into the lungs. The forced air leads to expansion of the 2 lungs with a positional orientation of the heart and hence twisting and obliteration of the ductus arteiosus forming the ligamentum arteriosum. Sometimes this duct may abnormally persist as what is called persistent ductus arteriosus.

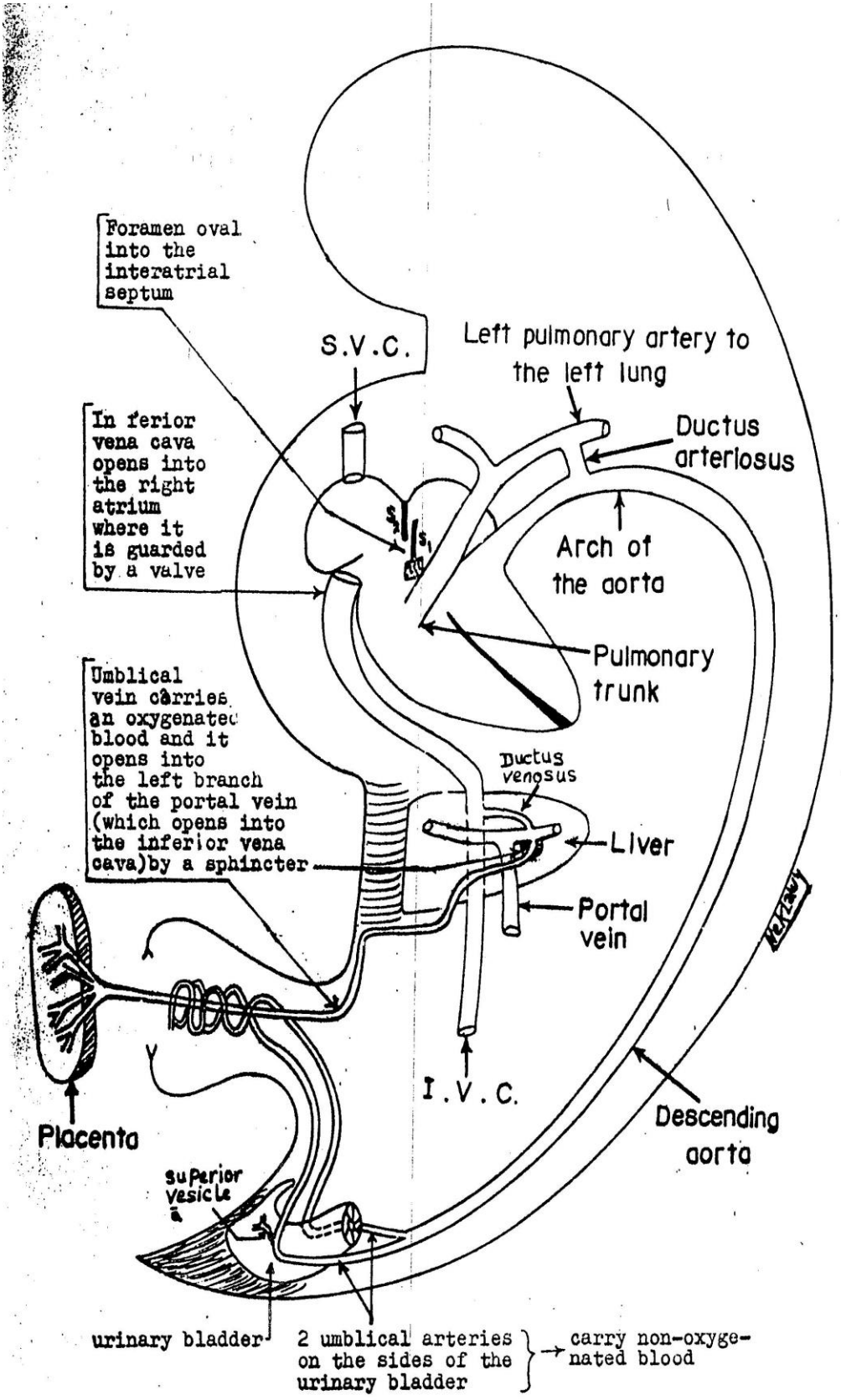
b- Thus, no venous blood circulates into the descending aorta and its two umbilical arteries(each gives a superior vesical artery) .A condition which leads to obliteration and fibrosis of the distal parts of these arteries (down to the level of their superior vesical branches) forming the two medial umbilical ligaments.

c- As the lungs are ready to function, the umbilical vein becomes of no role in carrying the oxygenated blood from the placenta resulting in its obliteration with the formation of the round ligament or ligamentum teres of the liver.

d- Similarly the ductus venosus is also, obliterated and fibrosed to form the ligamentum venosus.

e- As the venous blood is, forced into the lungs and returned back as an arterial blood towards the left atrium, the pressure into this atrium will increase and hence closure of the foramen oval by pressing the movable, flap- like, septum primum on the septum secondum. By closure of the foramen oval a depression on the righ

surface of the septum primum (fossa ovalis) will be left and being bounded above by the lower semilunar border of the septum secondum (annulus ovalis)


Foramen oval into the interatrial septum
S.V.C.
Left pulmonary artery to the left lung
Ductus arteriosus
Inferior vena cava opens into the right atrium where it is guarded by a valve
Arch of the aorta
Pulmonary trunk
Umblical vein carries an oxygenated blood and it opens into the left branch of the portal vein (which opens into the inferior vena cava) by a sphincter
Ductus venosus
Liver
Portal vein
I.V.C.
Placenta
Descending aorta
superior vesicle
urinary bladder
2 umblical arteries on the sides of the urinary bladder
carry non-oxygenated blood

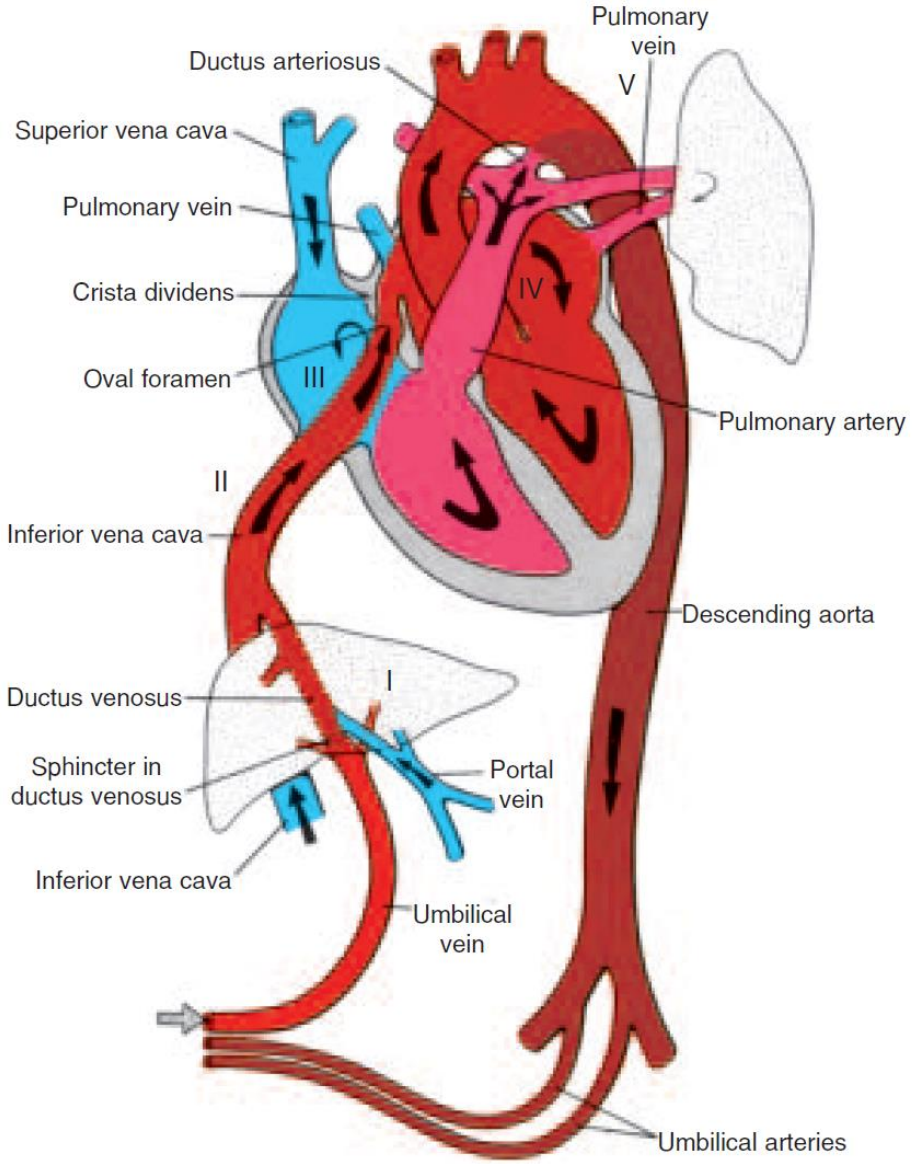


Figure 11.47 Fetal circulation before birth. *Arrows*, direction of blood flow. Note where oxygenated blood mixes with deoxygenated blood: in the liver (*I*), in the inferior vena cava (*II*), in the right atrium (*III*), in the left atrium (*IV*), and at the entrance of the ductus arteriosus into the descending aorta (*V*).

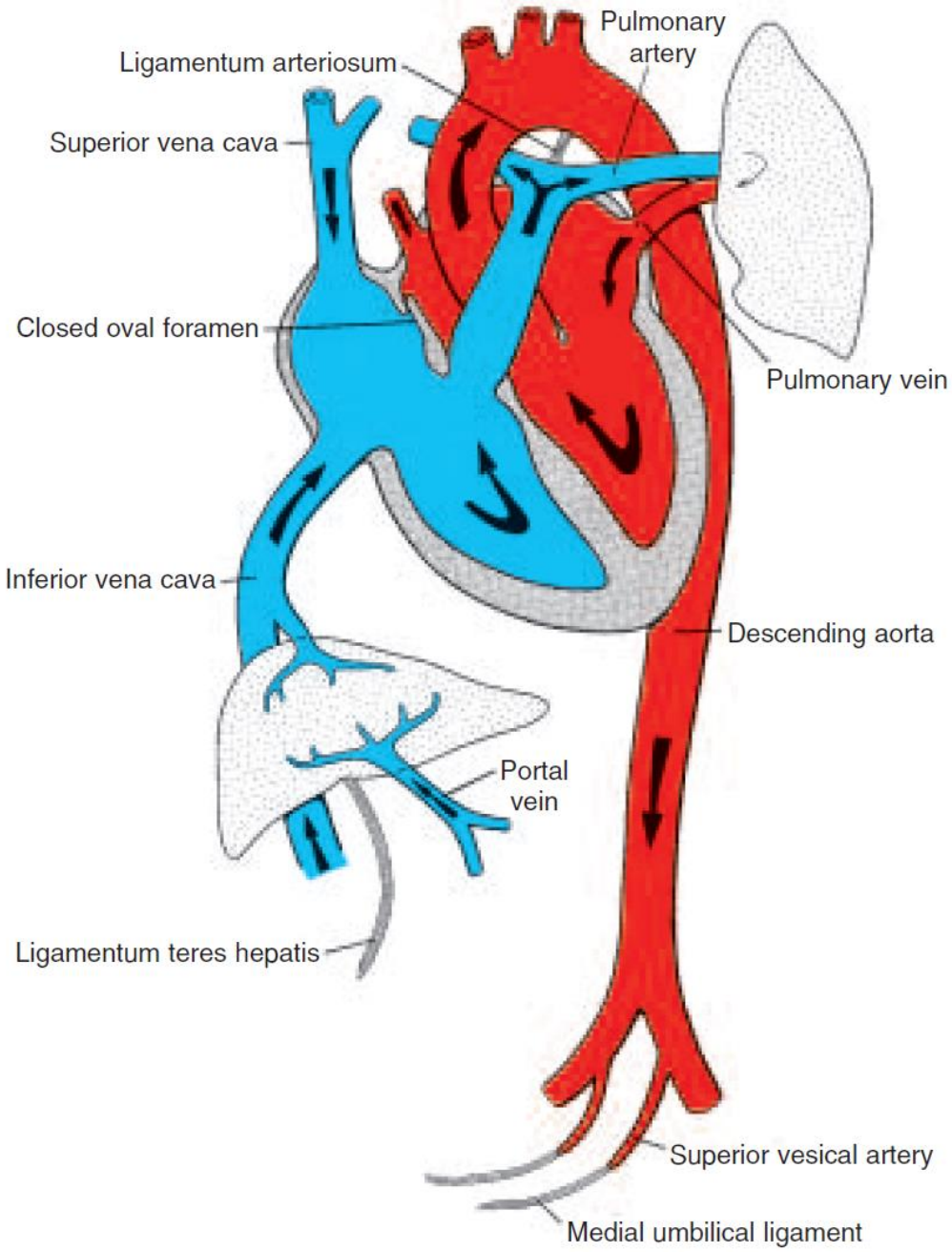


Figure 11.48 Human circulation after birth. Note the changes occurring as a result of the beginning of respiration and interruption of placental blood flow. *Arrows*, direction of blood flow.